

Excluding the exact- $\delta = 4$ stratum

in a quartic three-dimensional Keller leading family

Independent AI-assisted research program

First release: 26 July 2026 (UTC)

Abstract

Let $F : \mathbb{A}_{\mathbb{C}}^3 \rightarrow \mathbb{A}_{\mathbb{C}}^3$ be a polynomial map of total degree four with nonzero constant Jacobian. We consider the frozen fixed-quadratic line-double-cover leading family

$$H_4 = (h(p, q)p^2, h(p, q)q^2, 0),$$

where h is a binary quadratic. If $R = (H_3)_3$, set

$$\alpha = J(Q, R), \quad \beta = -J(P, R), \quad \gamma = J(P, Q), \quad \delta = \deg \gcd(\alpha, \beta, \gamma).$$

We prove that no Keller counterexample in this family has $\delta = 4$. A Hilbert–Burch and incidence calculation gives exactly six normal-form orbits at $\delta = 4$. Full lower-term determinant descents exclude all six, with specialization-safe boundary charts. The result is a structural exclusion inside one frozen quartic row. It does not prove a universal quartic exclusion and does not improve the known total-degree floor four.

Verification and review status. The proof and programs were produced with substantial language-model assistance. The work is not peer reviewed. Exact symbolic checks certify the encoded algebra, not the correctness of an unreviewed mathematical model or worldwide priority.

1 Scope and weighted determinant equations

Write a degree-four map, after subtracting its constant term, as

$$F = L(p, q, r)^t + H_2 + H_3 + H_4,$$

where $L \in \mathrm{GL}_3(\mathbb{C})$ and H_i is homogeneous of ordinary degree i . Introduce

$$\Delta(w) = \det \left(L + wJH_2 + w^2JH_3 + w^3JH_4 \right) = \sum_{j=0}^9 E_j w^j.$$

The Keller condition is equivalent to

$$E_0 = \det L \neq 0, \quad E_j = 0 \quad (1 \leq j \leq 9). \tag{1}$$

We work only in the binary fixed-quadratic line-double-cover row

$$H_4 = (P, Q, 0), \quad P = hp^2, \quad Q = hq^2, \quad h \in \mathbb{C}[p, q]_2, \tag{2}$$

and write $R = (H_3)_3 \in \mathbb{C}[p, q]_3$. The top identity $E_7 = 0$ is controlled by the binary minors

$$\alpha = J(Q, R), \quad \beta = -J(P, R), \quad \gamma = J(P, Q). \tag{3}$$

Here J denotes the two-variable Jacobian in p, q .

Theorem 1. *In the family (2), there is no quartic Keller counterexample for which*

$$\delta := \deg \gcd(\alpha, \beta, \gamma) = 4.$$

Remark 2. *The theorem concerns one leading-shape stratum. Other strata in the same fixed-quadratic row remain open. In particular, Theorem 1 does not imply that every dimension-three Keller counterexample has total degree at least five. The existing universal floor four is due to the degree-three theorem known from Moh–Sathaye, for which Vistoli gave a short conceptual proof [2].*

2 Why exactly six families occur

Put $g = \gcd(\alpha, \beta, \gamma)$ and $d = 5 - \delta$. After dividing by g , the three generator degrees are

$$(d, d, d + 1). \quad (4)$$

The following bound makes the high-incidence enumeration finite.

Lemma 3 (Hilbert–Burch deficit bound). *Assume the reduced minor row has height two and its first two entries are constant-linearly independent. If e_1, e_2 are its two minimal syzygy degrees, then*

$$e_1 + e_2 = 3d + 1.$$

The two gradient columns of (P, Q, R) give syzygies of degree $d + 3$. If $k_i = d + 3 - e_i$, then

$$0 \leq k_i \leq 2, \quad k_1 + k_2 = \delta. \quad (5)$$

Consequently $\delta \leq 4$; when $\delta = 4$, one has $(k_1, k_2) = (2, 2)$.

Proof. Hilbert–Burch applied to a height-two row of degrees $(d, d, d + 1)$ gives $e_1 + e_2 = 3d + 1$. Express the two gradient syzygies in a minimal syzygy basis. Their 2×2 coefficient determinant is a nonzero scalar multiple of g : wedging the gradient columns gives (α, β, γ) , while wedging a minimal basis gives the reduced row. The coefficient degrees are $k_i = d + 3 - e_i$. They are nonnegative, their sum is $\deg g = \delta$, and the degree pattern in (4) bounds each by two. This proves (5) and the conclusions. $\square$

Since

$$\gamma = J(hp^2, hq^2) = 8h^2pq, \quad (6)$$

the divisor g is selected from a finite list of exponent vectors on the roots of h , together with $p = 0$ and $q = 0$. Divisibility of $J(Q, R)$ and $J(P, R)$ gives a finite linear system for the four coefficients of R . Solving those systems and quotienting by the stabilizer of the double cover gives the following proposition.

Proposition 4 (Exact- $\delta = 4$ incidence atlas). *Up to the frozen cover stabilizer, the independent exact- $\delta = 4$ locus in (2) is the disjoint union of the following six orbits.*

ID	Normal form or modulus	g
D4-SF-21C	$h = XY, R = X^2Y, \kappa = -16/5$	pX^2Y
D4-SF-20CC	squarefree h , two fixed contacts, $\kappa = 16/5$	pqX^2
D4-SF-11CC	squarefree $h, R = h(p + q), \kappa = 16$	pqh
D4-DN-3	$h = L^2, R = L^3, L = p + q$	L^4
D4-DN-2C	$h = L^2, R = L^2(p - 2q), L = p + q$	pL^3
D4-DN-1CC	$h = L^2, R = L(2p^2 + pq + 2q^2), L = p + q$	pqL^2

Proof. Factor h either as two distinct roots or as L^2 . Equation (6) bounds the exponent of every possible prime factor of g . For each bounded exponent vector of total degree four, impose

$$g \mid J(Q, R), \quad g \mid J(P, R) \quad (7)$$

on a general binary cubic R . Empty systems are discarded; systems whose actual gcd is larger or smaller are routed to the corresponding exact stratum. In the doubled-root chart, (7) leaves the three displayed normal forms. In the squarefree chart, the reciprocal and branch-swap symmetries identify the surviving parameter values into the three displayed orbits. The exact enumeration checks all bounded exponent vectors and all exit values before quotienting.

Two independent ledgers implement this calculation. The primary ledger has $17 + 6 + 1$ high-incidence entries; a blinded reconstruction refines the exact- $\delta = 3$ part and gives the canonical $19 + 6 + 1$ ledger. They agree on all six exact- $\delta = 4$ orbits. The machine-readable denominators, boundary routing, and strict reconciliation checks accompany this paper. $\square$

3 The six lower-term exclusions

For every orbit in Proposition 4, the calculation begins with the complete E_7 kernel, forms the full E_6 system with every lower coefficient present, and covers each rank or pivot boundary separately. The terminal mechanisms are summarized below.

Family	Complete descent
D4-SF-21C	The E_6 contact locus is a plane. A rank-seven open chart and its rank-six directions fail at E_5 ; the rank-five origin loses all nonbinary nonlinear coefficients and exits through a plane Keller map.
D4-SF-20CC	The r^2 -contact coordinates vanish and E_6 leaves one contact line. Its nonzero chart fails at E_5 ; the origin collapses at E_4 and exits through a plane Keller map.
D4-SF-11CC	The complete contact plane splits into specialization-safe rank charts. The nonzero charts fail in E_5, E_4 ; the zero chart gives the same unconditional plane exit.
D4-DN-3	The contact radical is two conjugate planes. Their interiors fail at E_5 , their punctured intersection forces $\det L = 0$, and the origin collapses to the plane exit.
D4-DN-2C	The contact radical is again two conjugate planes. Their interiors fail at E_5 . On the punctured intersection, the full E_5 residual ideal splits into two branches; one forces $\det L = 0$, while the other has a nonzero E_3 residual. The origin collapses to the plane exit.
D4-DN-1CC	E_6 leaves one contact line. Its nonzero chart is inconsistent at E_4 ; at zero contact every nonlinear term is binary and the plane exit applies.

For completeness, we record the decisive identities in the last new family. On either transverse plane, with $\eta^2 = -2$, two lower-variable-free E_5 coefficients are

$$\frac{(2k + 3s)Q_1(k, s)}{162}, \quad \frac{(2k + 3s)Q_2(k, s)}{243}, \quad (8)$$

and an exact Bezout identity shows that $Q_1(1, t), Q_2(1, t)$ are coprime. Their common line is $2k + 3s = 0$. On its punctured part $k \neq 0$, the full E_5 residual ideal is

$$\langle \mathcal{Q}, \mathcal{AB} \rangle. \quad (9)$$

The r^2 -part of E_4 forces $\mathcal{S} = 0$, after which $\mathcal{Q} = \mathcal{D}^2$. On $\mathcal{B} = 0$, the remaining compatibility is $\mathcal{A}\mathcal{F}_B = 0$; the $\mathcal{F}_B = 0$ branch has $\det L = 0$, and $\mathcal{A} = 0$ is the other branch. There,

$$\det L = \frac{\mathcal{F}_A \mathcal{H}_A}{1152}. \quad (10)$$

If $\mathcal{F}_A \neq 0$, two E_3 equations have pivot

$$\frac{k^2 \mathcal{F}_A^2}{144},$$

while each remaining equation reduces to

$$\frac{k \mathcal{F}_A^2}{288} \neq 0. \quad (11)$$

At the origin, a fresh constant E_6 pivot gives

$$[p^3 r]E_4 = -3b_{qr}^2, \quad [q^3 r]E_4 = \frac{2}{3}(3b_{qr} + 2\ell_8)^2. \quad (12)$$

Thus all six nonbinary quadratic coefficients vanish.

Whenever a zero-contact chart leaves only binary nonlinear terms, a linear target normalization separates a two-variable Keller map of degree at most four and a triangular r -coordinate. Moh's unconditional plane theorem for degree below 100 [1] makes that plane map an automorphism. This is a proved bounded-degree theorem, not an assumption of the open plane Jacobian conjecture.

Proof of Theorem 1. Proposition 4 is exhaustive and disjoint. The complete lower-term descents summarized above exclude each of its six terminal orbits. Therefore no exact- $\delta = 4$ leading point in (2) can belong to a quartic Keller counterexample. $\square$

4 Exact certificates and limitations

The aggregate verifier performs the following checks.

1. It compares a six-entry proof manifest with the canonical blinded $19 + 6 + 1$ denominator.
2. It deletes one family and requires the bridge verification to fail.
3. It reruns the primary enumeration, blinded denominator, reconciliation, and frozen hashes.
4. It obtains the execution plan from the validated manifest and runs six distinct fail-closed family wrappers.
5. Each family wrapper requires exact terminal markers from symbolic algebra and an independently implemented replay, and contains deliberate coefficient mutations that must fail.
6. A separate hostile assembly wrapper reruns the complete aggregate, verifies that the containing global row remains open, and rejects mutations of the canonical ID, alias, proof path, terminal marker, denominator source, theorem scope, and advertised counts.

The terminal assembly marker is

EXACT_DELTA4_SIX_FAMILY_EXCLUSION_STRICT_PASS.

The hostile assembly terminates with

EXACT_DELTA4_HOSTILE_UMBRELLA_AUDIT_STRICT_PASS.

These checks are reproducible evidence for the stated stratum theorem. They do not constitute peer review. In particular, they do not close the remaining exact- $\delta = 3$, lower-incidence, or power-fibre families, the parent fixed-quadratic row, or the global quartic denominator.

References

- [1] R. Biggers, T. T. Moh, and M. Fried, *On the Jacobian conjecture and the configurations of roots*, J. Reine Angew. Math. **340** (1983), 140–213.
- [2] A. Vistoli, *The Jacobian conjecture in dimension 3 and degree 3*, J. Pure Appl. Algebra **142** (1999), 79–89.